

A Compact High Gain Dual Band Square Slotted Octagonal Patch Antenna for X, Ku and Upper 5G Bands

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Abstract— A compact microstrip line-fed dual-band antenna featuring an octagonal radiating patch with square-shaped slots and a parasitic element on a fractional ground plane is proposed for X-band, Ku-band, and upper 5G applications. The antenna is implemented on a Rogers RT5880 substrate with compact dimensions of $20 \times 20 \times 0.79 \text{ mm}^3$. It attains a maximum gain of 6.11 dB and an average radiation efficiency exceeding 93%, wherein the slot-loaded patch and ground-plane parasitic component collectively enhance bandwidth, improve gain, and ensure superior impedance matching. The design further exhibits excellent return loss at resonant frequencies of 13.036 GHz and 28.132 GHz, providing wide operating bandwidths that support high data rates and reduced latency across a broad spectrum encompassing working frequencies.

Keywords—Wideband antenna, X-Band, Ku-Band, upper 5G, mm-Wave, Rogers RT 5880, Octagonal Radiating Patch.

I. INTRODUCTION

The explosive growth of modern wireless communication systems spanning satellite communications, radar sensing, and emerging fifth-generation (5G) networks has fueled an increasing demand for compact, high-performance antennas capable of supporting wide bandwidths, multi-band operation, and high data rates [1]. In particular, applications that function within the X-band (8–12 GHz), Ku-band (12–18 GHz), and the millimeter-wave (mm-Wave) frequency range for upper 5G (around 28 GHz) require antennas that not only offer broad bandwidth but also maintain high gain, low return loss, and tremendous radiation efficacy within a compact physical footprint [2]. These requirements are critical for enabling seamless high-speed connectivity, real-time data transmission, and efficient spectral utilisation in next-generation wireless systems. Microstrip patch antennas (MPAs) are well-recognised in this context for their low profile, simplicity of manufacture, and compatibility with microchips, making them suitable for applications ranging from satellite transceivers to handheld 5G devices [3]. However, despite substantial research efforts, many reported

designs still face trade-offs among size, gain, bandwidth, and radiation efficiency, limiting their applicability in multi-band and high-frequency scenarios. For instance, some designs achieve wide bandwidth but require large substrate dimensions, while others prioritize compactness at the expense of gain or impedance matching.

A substantial quantity of aeriels in the mm-Wave band, as well as in bands such as X-band and Ku-band, have been developed; nevertheless, many exhibit limitations in either bandwidth, gain, or efficiency. In [4], the researchers presented a $35 \times 33 \times 1.6 \text{ mm}^3$ crescent-shaped patch antenna designed for multi-band applications. The dimensions of this crescent-shaped antenna are relatively enormous, exhibiting an 80% radiation efficiency. The maximum gain of this antenna is 5.8 dB. A compact wideband fractal antenna suitable for WiMAX, WLAN, and C and X band uses is presented in [5], exhibiting a gain of 9 dB. Still, the return loss values at various resonant frequencies are relatively lower. The antenna's radiation efficiency is 92%. A compact MPA designed for Ku-band is presented in [6], exhibiting a gain of 6.31 dB and a reflection coefficient of -26.55 dB. The antenna demonstrates a radiation efficiency of 98.99%. In [7], the authors developed a microstrip patch antenna with somewhat higher dimensions ($40 \times 48 \times 1.59 \text{ mm}^3$) in the Ku frequency range for satellite uses. The prototype demonstrates a minimal peak gain. The return loss values are -17 dB and -25 dB, indicating inferior performance. A Slotted Ground Plane wideband MPA for 5G applications is presented in [8]. The engineered prototype demonstrated 70% radiation efficiency with a maximum gain of 5.22 dB. The return loss value is exceptional, almost -35 dB. A $40 \times 40 \times 1.6 \text{ mm}^3$ microstrip patch antenna in the shape of a heart is presented in [9] and functions in the 9.2–9.5 GHz frequency range. The constructed framework yielded an enhancement of 2.3 dB. The radiation efficiency of this antenna is at 42.22%, engineered explicitly for Search and Rescue Transponder (SART) systems. A slotted patch microstrip line feed antenna utilising Rogers RT5880 material for 28 GHz 5G applications

is presented in [10]. The constructed structure is a high-antenna with a notable radiation efficiency of approximately 89.7%. The total dimensions of this antenna are $40 \times 30 \times 1.575$ mm³. A compact wideband rectangular MPA featuring a fractional ground plane designed for 5G communication uses is presented in [11]. The total bandwidth of this design is 2.56 GHz. The gain is 8.39 dB, and the radiation efficiency is 84.5%. A flawed ground structure-based MPA exhibiting relatively poor radiation efficiency, around 66%, is reported in [12]. The gain of this configuration is 5.11 dB. The antenna is diminutive in dimensions. A microstrip patch wideband antenna including a double-T-shaped radiating patch and an altered ground plane is shown for 5G applications in [13]. The antenna offers substantial bandwidth, gain, and radiation efficiency. The dimensions of this design are relatively significant.

Despite notable progress, existing antennas targeting X-band, Ku-band, and upper 5G systems still face significant limitations. Many designs offer wide bandwidth but require large dimensions [4], while others maintain compact profiles but suffer from reduced gain and radiation efficiency [5]–[7]. Designs optimized for mmWave 5G either operate only at a single band [8], [10], [11] or fail to maintain adequate efficiency, especially with slotted ground configurations [12]. Furthermore, some compact models demonstrate acceptable performance but lack multi-band support across X-band, Ku-band, and 28 GHz simultaneously [9], [13]. There remains a need for a compact dual-band antenna that simultaneously provides wide bandwidth, high radiation efficiency, good impedance matching, and moderate gain across X-band, Ku-band, and upper 5G frequencies. To address this, the present work proposes a compact microstrip line-fed octagonal patch antenna incorporating square slots and a parasitic element on a fractional ground plane. This dual-band configuration enhances bandwidth, gain, and impedance matching while maintaining a small footprint and demonstrates strong performance for X-band, Ku-band, and upper 5G applications, as validated through “CST Microwave Studio” simulation software. The aerial functions within the frequency range of 8.802 GHz to 15.182 GHz and 24.802 GHz to 32.727 GHz with the values of return loss -39.28 dB and -58.076 dB, respectively. The main radiating patch is constructed from annealed copper, utilizing Rogers RT 5880 (lossy) as the substrate. The substrate area of the offered antenna measures 20 mm by 20 mm. The VSWR value approaches 1. The mean radiation efficiency is 93%. The leftovers of the manuscript are structured as follows: The configuration of the engineered slotted antenna is elaborated upon in Section II. Subsequently, the analysis and discussion of the simulated results are presented in Section III, followed by the comparative evaluation in Section IV, future work in Section V, and the conclusion in Section VI.

II. PROPOSED DESIGN OF THE DUAL-BAND ANTENNA

A compact antenna component is demonstrated in Fig. 1, and the antenna component measurements are shown in Table I. The proposed single-element MPA is made of Rogers RT5880 substrate material. This material exhibits specific characteristics, including a thickness of 0.79 mm, a relative permittivity (ϵ_r) of 2.2, and a tangent loss ($\tan \delta$) of 0.0009. The substrate used to build the antenna element has dimensions of $20 \times 20 \times 0.79$ mm³. The antenna element comprises a combination of an octagon-shaped main radiating element and five square slots etched within the main radiating

element. The proposed slotted octagonal patch improves current distribution and enables dual-band operation without increasing size. A 10×2 mm² parasitic strip on the fractional ground enhances gain and bandwidth, especially at the upper 5G band, outperforming conventional DGS or full-ground designs [6], [10], [12]. The 0.035-mm copper layer forms the radiating patch, feed line, and ground. The square slots concentrate current paths for better impedance matching and the octagonal geometry ensures smoother current transitions, resulting in improved bandwidth.

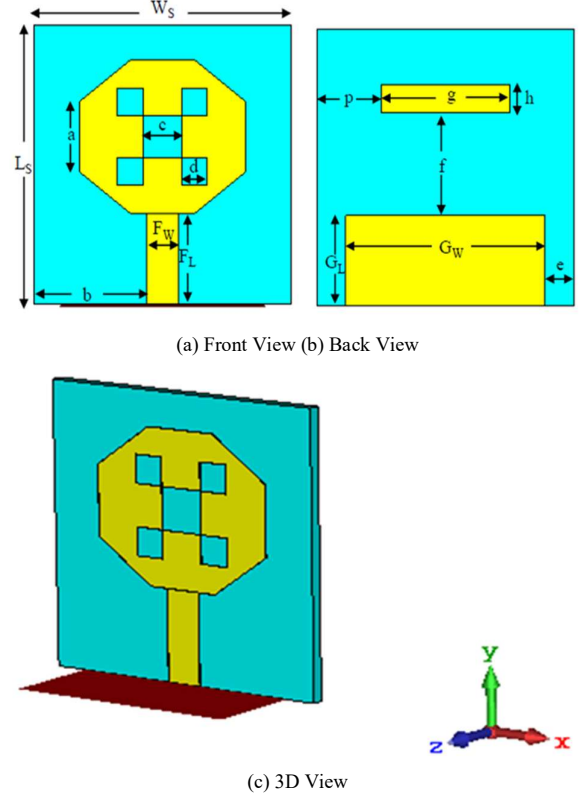


Fig. 1. Octagon-shaped MPA with parasitic element in ground plane.

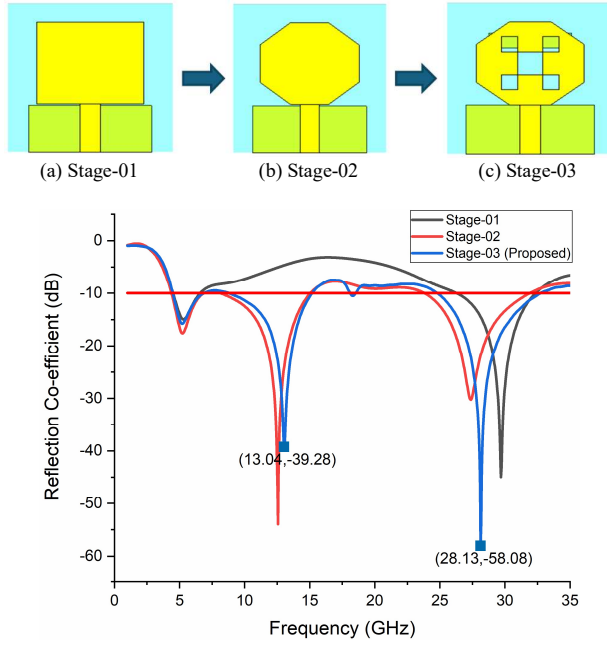
TABLE I. DESIGN SUMMARY OF THE PROPOSED MPA

Antenna Parameters	Length (mm)	Antenna Parameters	Length (mm)
$W_s = L_s$	20	G_L	6.50
a	05	G_w	15.50
b	8.77	e	2.25
c	03	f	7.5
d	02	g	10
F_w	2.45	h	2
F_L	6.50	p	5

III. SIMULATION RESULTS AND ANALYSIS OF THE PROPOSED ANTENNA

A. Reflection Co-efficient and VSWR analysis

Fig. 2 displays the design steps along with scattering parameters of the octagon-shaped MPA with a parasitic element in the ground plane. Initially, a quadrilateral-molded radiating patch with a fractional ground plane is made up of annealed copper. Then, from that rectangle, a regular octagon with a side length of 5 mm is formed. Finally, five square-shaped slots are etched from that octagon, and a parasitic component is introduced in the back plane for gain enhancement.



(d) S_{11} -parameter curve of the three design steps for the proposed MPA.

Fig. 2. Design steps (a-c) and corresponding S-parameters (d) of the proposed MPA.

The offered antenna is a dual-band antenna. It operates from 8.80 GHz to 15.18 GHz in the first region with a bandwidth (considered at -10 dB points) of 6.38 GHz. It provides excellent bandwidth coverage for both X-band and Ku-band uses. Both are used in satellite communication. The value of the reflection coefficient is -39.28 dB with a resonant frequency of 13.036 GHz. In the other operating zone, the antenna functions at a 28.132 GHz frequency with a reflection coefficient of -58.076 dB, making it appropriate for upper 5G applications. The bandwidth of the antenna in this band is 7.92 GHz, as it operates from 24.80 GHz to 32.72 GHz. The proposed MPA is an excellent candidate for both X-band and Ku-band uses, as well as for upper 5G applications.

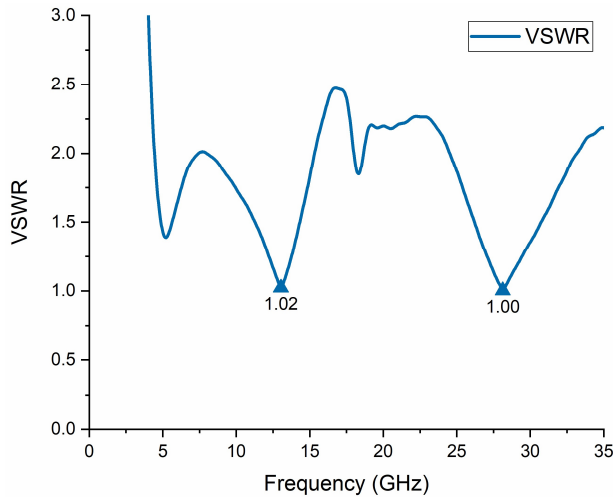


Fig. 3. VSWR of the proposed MPA.

The Voltage Standing Wave Ratio (VSWR) is an important aspect to consider for evaluating the quality of impedance matching. Fig. 3 displays the fluctuation of VSWR

over both the complete bands of the patch antenna. The VSWR over both the first and second bandwidths lies between 1 and 2 ($1 < \text{VSWR} < 2$). At the first resonant point, the magnitude of VSWR is 1.021, which is almost close to unity. In the second resonant point, the value of VSWR is 1.0025. In both operating bands, the values of VSWR are close to unity, ensuring good impedance matching for the proposed MPA. The concentrated port signal will be captured by the antenna and further emitted by it.

The relationship between antenna impedance and frequency is illustrated in Fig. 4, providing valuable insights into their interaction. At the first resonant point, the real part is 47.91Ω , which is very near to 50Ω , and the imaginary part is -0.596Ω , which is close to 0. At 13.036 GHz, the impedance is $47.91-j0.596 \Omega$. At 28.132 GHz, the value of impedance is $48.85-j0.097 \Omega$. So, at this frequency, the impedances are also close to the ideal values of 50Ω for the real part and 0 for the imaginary part. A comprehensive overview of the antenna operation is provided in Table II.

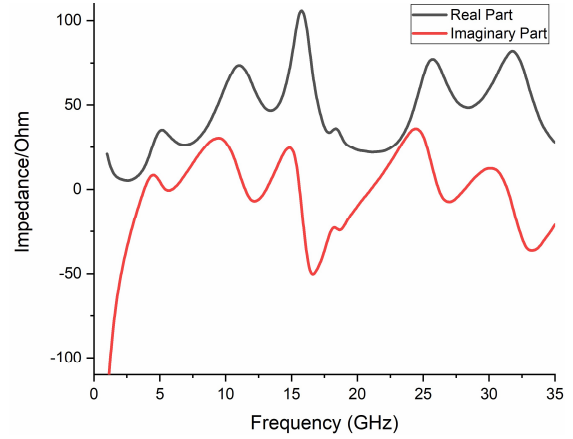


Fig. 4. Impedance Vs Frequency curve of the designed MPA both Real part and Imaginary part

B. Radiation Performance and Characteristics

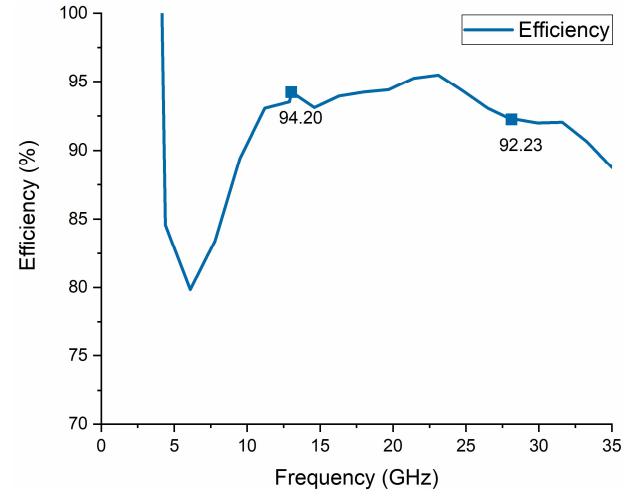


Fig. 5. Radiation Efficiency of the proposed MPA.

The curve of radiation efficacy (linear) vs frequency is depicted in Fig. 5. It is observed from the curve that, in the first operating region, the value of radiation efficiency lies between 89% and more than 94%. The maximum radiation efficiency is 94.2%, which is observed at the resonant point of

13.036 GHz. In the other operating region, the radiation efficiency exceeds 92%. At the resonant point, it reaches 92%, indicating that the proposed design also exhibits excellent radiation characteristics when implemented for upper 5G applications.

The three-dimensional gain and directivity of the system can be effectively analyzed and optimized for improved performance of the square slotted antenna in the first operating region are demonstrated in Fig. 6. It is observed from the Fig. 6 (a, b) that at resonance, the gain value is 3.76 dB and the directivity value is 4.02 dBi, respectively. The 3D gain and directivity of the antenna when it operates in the upper 5G region are illustrated in Fig. 7. At the resonant frequency of 28.132 GHz, the gain value is 6.11 dB and the directivity is 6.46 dBi respectively in Fig. 7 (a) and Fig. 7 (b). The linear graph of gain and directivity vs frequency is illustrated in Fig. 8. At the first operating band, the gain varies from 0.67 dB to 4.27 dB. The peak gain is observed at a frequency of 14.6 GHz. The directivity in this region varies from 3.50 dBi to 4.67 dBi. In the second operating region, the gain exceeds 4.50 dB. The directivity varies from 6.13 dBi to 6.66 dBi. The addition of a parasitic element and etched slots enhanced the gain at the final design step. So, the proposed design exhibits excellent gain and directivity in both bands.

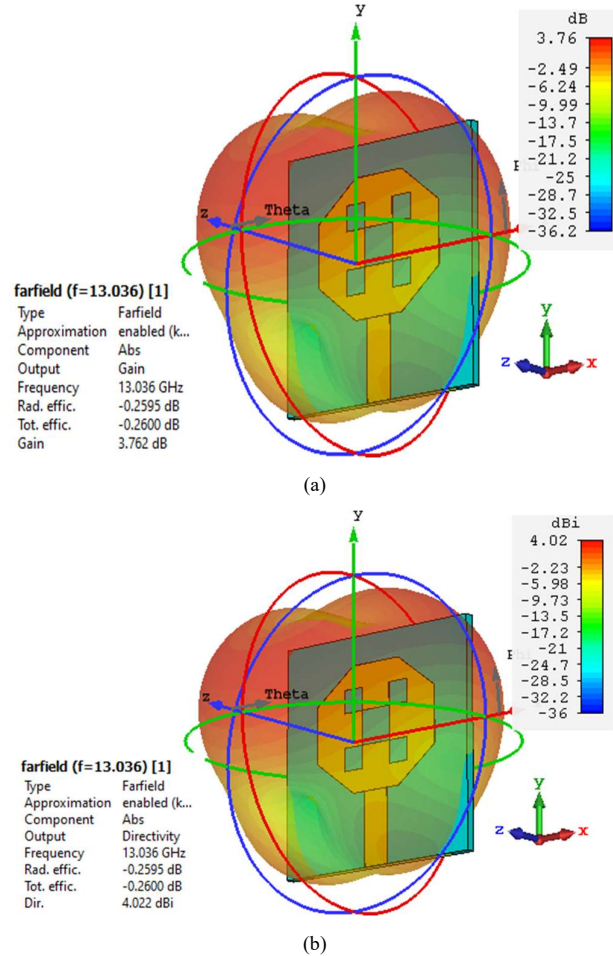


Fig. 6. Radiation pattern of 3D (a) Gain and (b) Directivity at 13.036 GHz.

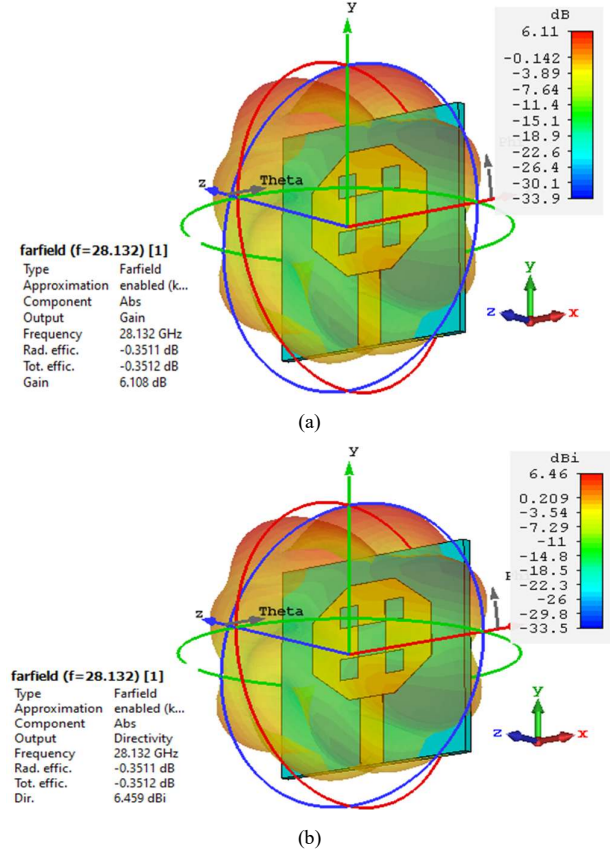


Fig. 7. Radiation pattern of 3D (a) Gain and (b) Directivity at 28.132 GHz.

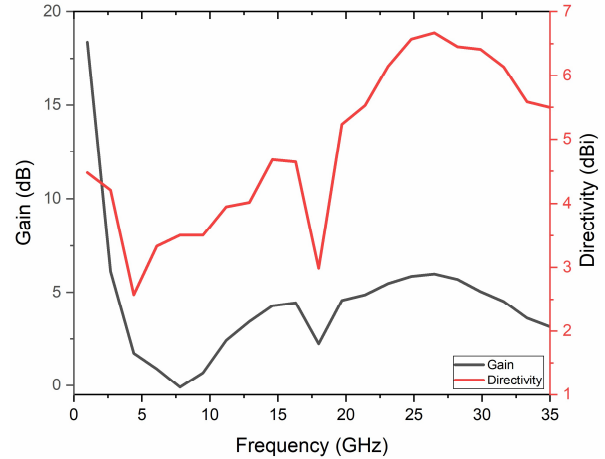


Fig. 8. Gain and Directivity vs Frequency.

The numerous polar strategies of radiation configurations (E- and H-field) of the offered antenna at 13.036 GHz and 28.132 GHz is offered in Fig. 9 and 10, respectively. From the polar scheme of the radiation configuration in Fig. 9, the main lobe is concentrated at 0° for $\phi=0^\circ$ and 160° for $\phi=90^\circ$. The main lobe direction at 0° indicates broadside radiation suitable for satellite-pointing applications. At 13.036 GHz half-power beamwidth of 82.8° (E-plane) and 64.1° (H-plane) provides moderately wide coverage that desirable for Ku-band transceivers. The main lobe level is 17.3 dBV/m for $\phi=0^\circ$ and 18.5 dBV/m for $\phi=90^\circ$ in the case of E-field. On the other hand, the main lobe level is -34.2 dBA/m for $\phi=0^\circ$ and -33 dBA/m for $\phi=90^\circ$ in the case of H-field. The side lobe level is -9 dB.

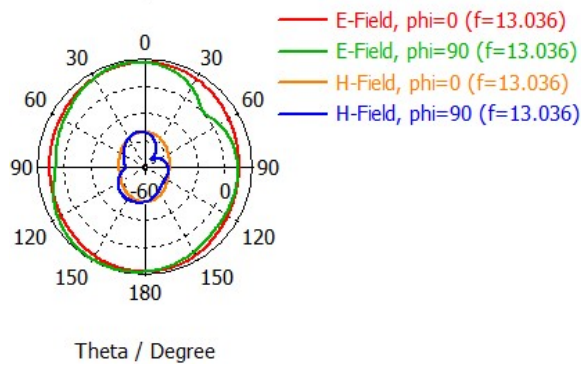


Fig. 9. E- and H-field at $\phi=0^\circ$ and $\phi=90^\circ$ at 13.036 GHz

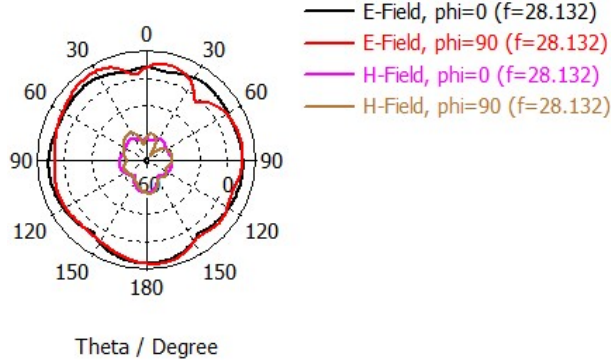
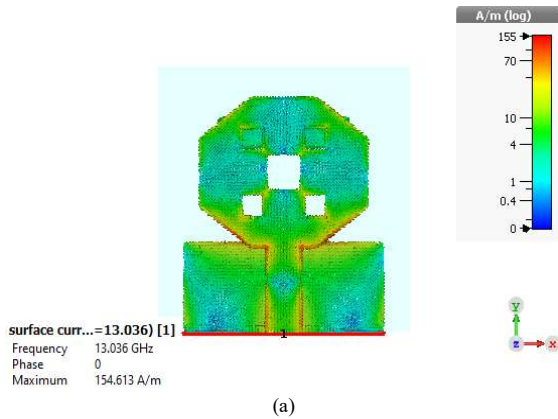


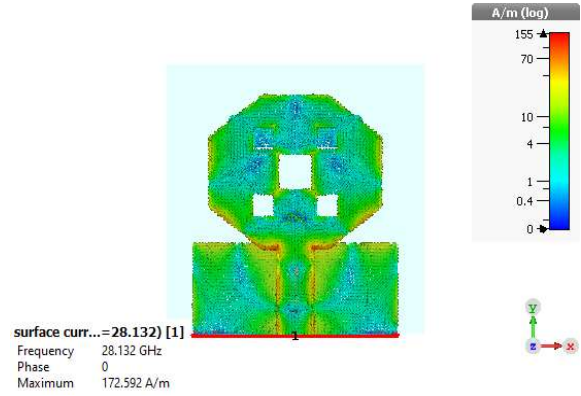
Fig. 10. E- and H-field at $\phi=0^\circ$ and $\phi=90^\circ$ at 28.132 GHz.

The E- and H-field patterns corresponding to 28.132 GHz are shown in Fig. 10. From the polar scheme of the radiation pattern, the main lobe is focused at 180° for $\phi=0^\circ$ and 173° for $\phi=90^\circ$. The half-power beam-width is 37.9° for $\phi=0^\circ$ and 33.1° for $\phi=90^\circ$. The narrower beamwidth of $33\text{--}38^\circ$ indicates improved directionality, beneficial for mmWave 5G links requiring higher spatial selectivity. The main lobe magnitude is 15.4 dBV/m for $\phi=0^\circ$ and 15.8 dBV/m for $\phi=90^\circ$ in the case of E-field. On the other hand, in the case of the H-field, the main lobe level is -36.2 dBA/m for $\phi=0^\circ$ and -35.7 dBA/m for $\phi=90^\circ$. The lowest side lobe magnitude is -4.3 dB.

C. Surface Current Distribution



(a)



(b)

Fig. 11. Surface Current of the proposed MPA at (a) 13.036 GHz and (b) 28.132 GHz.

The distribution of the surface current of the offered antenna at 13.036 GHz and 28.132 GHz is depicted in Fig. 11. The surface current densities are higher at the edges of the patch for both frequencies. At 13.036 GHz, the peak value of surface current is 154.613 A/m, while the peak surface current at 28.132 GHz is 172.592 A/m.

TABLE II. PERFORMANCE METRICS OF THE PROPOSED MPA

Performance Parameter	Value
Resonant Frequency (GHz)	13.036, 28.132
Lower Cut off Frequency (GHz)	8.802, 24.802
Higher Cut off Frequency (GHz)	15.182, 32.727
-10 dB Bandwidth (GHz)	6.38, 7.92
Return loss (dB)	-39.28, -58.076
Gain (dB)	4.27, 6.11
Directivity (dBi)	4.67, 6.66
VSWR	1.021, 1.0025
Average Radiation Efficiency (%)	93

IV. COMPARATIVE ANALYSIS

The proposed antenna outperforms others in regards to bandwidth and return loss, with high radiation efficiency and compact size of $(20 \times 20 \times 0.79 \text{ mm}^3)$. Compared to [4]–[13] in Table III, it offers wider frequency coverage (X-band, Ku-band, 5G) and competitive gain, making it highly suitable for modern wireless applications. No prior design in the literature simultaneously covers 8.8–15.18 GHz and 24.8–32.7 GHz within such a $20 \times 20 \text{ mm}^2$ footprint.

TABLE III. COMPARISON WITH RECENT RELEVANT WORKS

Ref.	Dimensions (L×W×h) (mm ³)	Operating frequency range (GHz)	Operation Bandwidth (GHz)	Centre frequency (GHz)	Return loss (dB)	Max. Gain (dB)	Radiation Efficiency (%)	Band applications
[4]	35×33×1.6	2.8 – 3.2, 6.16 – 10.45	0.4, 4.29	3.1, 9.3	-33 -32	5.8	>80	X- Band
[6]	14.1×18.6×1.6	11.595 – 15.75	4.155	12.54	-26.55	6.31	98.99	Ku Band
[7]	40×48×1.59	12.80 – 15.83	3.03	12.8, 14.9	-25, -17	3.67	N/A	Ku Band
[8]	20×20×0.79	26.154 – 31	4.846	27.47	-35.03	5.22	70	5G
[9]	40×40×1.6	9.18 – 9.63	0.45	9.35	-41.83	2.3	42.22	SART
[10]	40×30×1.575	24.356 – 29.197	4.841	25.69	-42.134	8.104	89.77	5G
[11]	20×17×1.575	26.984 – 29.551	2.567	28.201	-31.677	8.39	84.5	5G
[12]	20×20×0.79	25.52 – 31.37	5.85	27.26	-35.353	5.11	66	5G
[13]	30×30×1.575	24.333 – 31.801	7.468	26.88	-40.36	6.367	92	5G
This work	20×20×0.79	8.8029 – 15.182, 24.802 – 32.727	6.38, 7.92	13.036, 28.132	-39.28, -58.076	4.27, 6.11	93	X-band, Ku- band, 5G

V. FUTURE WORK

Although this work focuses on simulation-based analysis, the results indicate strong potential for fabrication due to simple single-layer geometry. To enhance the practical applicability of the proposed antenna in real-world wireless systems, future work will involve on fabricating and experimentally testing a physical prototype to validate simulated results. Additionally, the design may be extended to support MIMO configurations for higher channel capacity in 5G networks. Finally, performance analysis under various environmental conditions will be pursued.

VI. CONCLUSION

A compact octave-shaped dual-band microstrip antenna with square-slot loading and a ground-plane parasitic element has been designed and evaluated for X-band, Ku-band, and upper 5G operation. Through electromagnetic optimization, the proposed structure successfully achieves wide bandwidths (6.38 GHz and 7.92 GHz), excellent return loss (−39.28 dB and −58.076 dB), high radiation efficiency, and gains of 4.27 dB and 6.11 dB. The slot-loaded octagon and parasitic ground element collaboratively enhance impedance matching, efficiency, and directivity while maintaining a highly compact $20 \times 20 \text{ mm}^2$ footprint. These characteristics demonstrate that the antenna not only supports multi-band applications but also addresses key limitations of prior designs in gain, bandwidth, and efficiency. Future work will include fabrication and experimental validation to further demonstrate the practical feasibility of the proposed model in real-world X-band, Ku-band, and 5G systems.

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